Trigonometry in 3D

1

a

i $EG = 7\sqrt{2}$ cm   ii $AG = 7\sqrt{3}$ cm

b

i 45° ii 35° iii 90° iv 55°

2

a $z = 14.32$ cm   b $\theta = 11.83^\circ$

3

a $z = 21.47$ cm   b $75.15\ \text{cm}^2$

4

a 11.31° b $CE = \sqrt{325}$ c $BE = \sqrt{334}$ d 9.45°

e $CX = \sqrt{250}$ f $z = 10.74^\circ$

5

a $x = 15$ b $y = 17$ c $\theta = 28^\circ$

6

a

i $\alpha = 90^\circ$ ii $\beta = 90^\circ$ iii $\gamma = 90^\circ$

b $x = \sqrt{2}$

c $y = \sqrt{3}$

d $\theta = 35^\circ$

7

a $b = \sqrt{746}$ b $109\ \text{cm}^2$ c $\theta = 16^\circ$

8

a 91.92 m   b 37.14 m   c $11^\circ 25'$ d $15^\circ 57'$

9

a $BC = 82$ m   b 48 m

10 $\theta = 65^\circ 9'$

11

a $z = 8.49$ cm b $\theta = 68.34^\circ$

12

a $MD = \sqrt{32}$ cm b $x = 61.87^\circ$ c $h = \sqrt{112}$ cm d $MN = 4$ cm

e $PN = \sqrt{128}$ cm f 69.30° g 70.53°

13 $AF = 55.77$

14

a $h = 9.37$ m b $\theta = 25^\circ$

15

a $x = 56$ b $h = 90$

16

a $h = 47$ b $x = 131$ c $y = 97$

17

a $\overline{AE} = 26.0$ m b $\theta = 31^\circ 14'$

18

a 2481 m b 1486 m c 1384 m

19

a

i $VA = 8.49$ cm ii $VC = 7.81$ cm

iii $AC = 7.81$ cm

b $C = 65^\circ 51'$

20

a $CD = 22.65$ m b $AB = 18.29$ m c 16.18 m

21 41 m

22

a $d = \sqrt{41}$ m b $h = 9.86$ m c $x = 63.11^\circ$ d $y = 67.92^\circ$

23

a 1282 m b 2033 m c $AB = 1977$ m